

Question			Marking details	Marks available					
				AO1	AO2	AO3	Total	Maths	Prac
		(iii)	Increasing v increases redshift OR linking orbital speed to redshift (can be implied by good answers) (1) Correct realisation that factor is $\sqrt{5}$ or that v^2 proportional to force (this implies the 1st mark) (1) Joseff is wrong (based on some valid physics) (1) e.g. red shift increases with orbital speed but they aren't proportional (force and speed) so Jo is wrong (this gets 2 marks – 1st and 3rd) Also accept for 2 marks: Red shift increases with orbital speed and orbital speed is proportional to force, so Jo is right Also accept calculations: Expected $v = 144\,000\text{ m s}^{-1}$ (1) Red shift = 0.21 n[m] (1) Joseff is wrong (1)			3	3	2	
	(e)		[More] experiments involving Higgs [decay] (1) Results [need to be] in agreement with theory (1)			2	2		
			Question 5 total	8	9	5	22	12	0